Appendix of Beyond Surface Cues: Multimodal Implicit Emotion Understanding with a New Dataset, Reasoning Framework, and Benchmark

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A Dataset Construction and Annotation Details

A.1 Representative Examples of the 13 Emotion Labels in MIE

Table 1 presents representative examples of the 13 emotion labels in MIE, covering seven explicit categories and six implicit categories. By placing explicit and implicit examples side by side under a unified taxonomy, the table provides an intuitive illustration of the core setting of MIE: emotion understanding in realistic multimodal scenes involves not only recognizing *which* emotion is present, but also determining *how* that emotion is expressed. The examples show that explicit and implicit labels are not separated by category identity alone, but by the relationship between observable behavior and the underlying affective state. This makes the table a useful visual summary of the dataset design and the motivation for explicit–implicit joint modeling.

A clear contrast can be observed between the two expression forms. Explicit examples typically exhibit relatively salient affective cues, such as visible facial changes, strong behavioral reactions, or direct emotional expressiveness, making their labels more readily inferable from surface appearance. In contrast, implicit examples are often more restrained, ambiguous, or behaviorally indirect: the underlying emotional state is not directly readable from facial expression alone, and may even appear partially inconsistent with the surface behavior. This side-by-side comparison highlights why implicit emotion understanding is substantially more challenging than explicit recognition. The difficulty does not simply come from weaker emotion intensity, but from the fact that the true affective state must often be recovered from subtle multimodal traces and local situational meaning rather than overt emotional display.

More importantly, the representative examples illustrate the practical significance of the MIE task formulation. In real interactions, a system must first judge whether the current emotion is directly expressed or implicitly conveyed, and only then determine the appropriate emotion category. The examples therefore reinforce that MIE is not a conventional emotion classification benchmark with more labels, but a dataset specifically designed to evaluate joint reasoning over emotion category and expression form. In this sense, the table does more than provide qualitative illustration: it visually clarifies the gap between observable emotional surface and latent affective state, and thereby explains why multimodal evidence organization and context-aware reasoning are essential for strong performance on MIE.

A.2 Alignment Between Annotation Segments and Emotion Judgment Segments

Table 2 presents representative pairs of *annotation segments* and *emotion judgment segments* for the implicit emotion categories in MIE, illustrating how contextual evidence is used to support reliable annotation. A clear pattern is that the annotation segment itself often contains only weak, restrained, or behaviorally indirect affective signals, making the underlying emotion difficult to determine from the target clip alone. By contrast, the corresponding emotion judgment segment provides a more interpretable local context in which the latent affective state becomes easier to verify. This side-by-side comparison shows that implicit emotion annotation in MIE is not based on arbitrary speculation, but on context-grounded emotional validation.

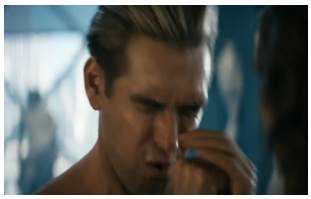
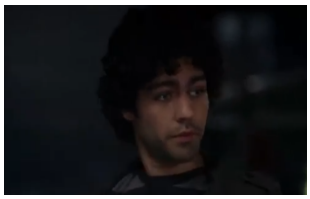
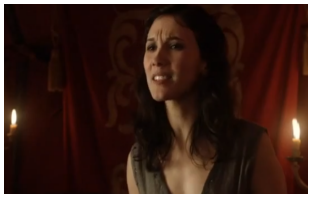
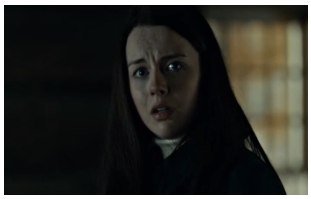
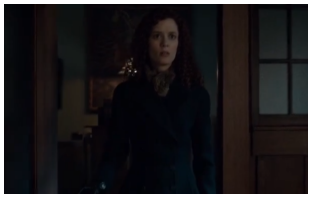
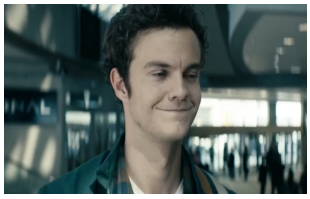
More importantly, the table highlights the distinction between *where the label is assigned* and *where the emotional evidence is verified*. The final label is still assigned to the annotation segment itself, since that segment is the actual target of modeling and evaluation. However, the paired emotion judgment segment helps annotators determine whether the subtle behavior observed in the target clip indeed corresponds to the intended latent emotion. For example, in several cases the annotation segment appears emotionally restrained or even superficially neutral, whereas the judgment segment reveals a clearer affective tendency through subsequent facial, behavioral, or situational development. This design is particularly important for implicit emotions, whose true states are often only partially reflected in the target clip and therefore require controlled contextual grounding during annotation.

Overall, these examples demonstrate that the alignment between annotation segments and emotion judgment segments is both *semantic* and *functional*. Semantically, the two segments refer to the same underlying emotional state within a coherent local narrative context; functionally, they play different roles in dataset construction, with the former serving as the labeling target and the latter serving as contextual validation evidence. This design helps reduce annotation ambiguity, improves label reliability, and preserves the intended difficulty of the benchmark. In other words, context is used not to replace the target segment, but to ensure that the assigned implicit label is well grounded rather than weakly inferred from surface cues alone.

A.3 Category Definitions and Annotation Criteria

Table 3 presents the hierarchical emotion taxonomy adopted in MIE. The taxonomy is designed according to three complementary principles: semantic coherence, annotation operability, and modeling suitability. First, each main category is defined as a coarse-grained affective state with a clear psychological core, so that semantically related fine-grained emotions can be grouped under a shared higher-level concept. For example, *Love* covers attachment-, desire-, and expectation-related positive states, whereas *Joy* emphasizes pleasure, fulfillment, and elevated motivation. Similarly, *Anger*, *Sadness*, and *Fear* are separated not only by valence, but also by their distinct affective functions, such as outward frustration and aversion,

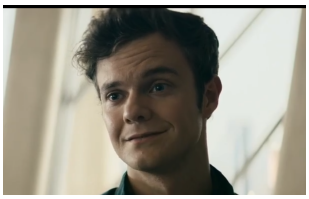
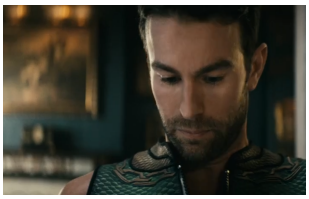
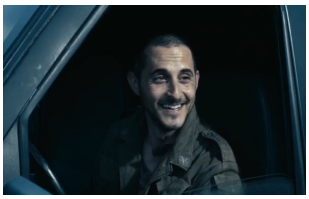
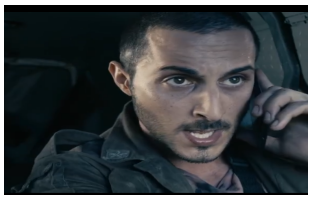
Table 1: Example illustrations for the 13 emotion labels in MIE.

Emotion Label	Explicit	Implicit
Neutral		
Love		
Joy		
Surprise		
Anger		
Sadness		
Fear		

inward loss and deprivation, or threat- and uncertainty-related anticipation. This hierarchical design reduces excessive label fragmentation while preserving meaningful emotional distinctions.

Second, the taxonomy is constructed to support reliable human annotation in realistic multimodal scenes. Instead of directly annotating a large number of highly fine-grained labels, MIE uses

Table 2: Illustrative examples of annotation segments and emotion judgment segments for the 13 emotion labels in MIE.

Emotion Label	Annotation Segment	Emotion Judgment Segment
Imp-Love		
Imp-Joy		
Imp-Surprise		
Imp-Anger		
Imp-Sadness		
Imp-Fear		

the fine-grained emotions as representative semantic references and maps them into seven main categories. This strategy improves label consistency because annotators can rely on prototypical affective meanings while avoiding overly subtle distinctions that are difficult to verify from multimodal evidence alone. In particular, *Surprise* is treated as a separate mixed-valence category because its emotional polarity is highly context-dependent, and *Neutral* is retained as a distinct non-salient affective state to account for scenes where no clear emotion is directly expressed. The inclusion of *Representative Meaning*, *Fine-Grained Emotions*, and *Valence* in the taxonomy therefore provides both conceptual guidance and practical annotation support.

Finally, this taxonomy is also suitable for the explicit–implicit joint modeling setting of MIE. The seven-category main taxonomy provides a unified semantic space for emotion recognition, while the distinction between explicit and implicit expression is handled as an additional layer of affective interpretation rather than by redefining the emotion categories themselves. As a result, the model is required not only to recognize *what* emotion is present, but also to determine *how* it is expressed. In this sense, the proposed taxonomy

Table 3: Hierarchical emotion taxonomy used in MIE.

Main Category	Representative Meaning	Fine-Grained Emotions	Valence
Neutral	Emotionally stable or non-salient affective state	Neutral	Neutral
Love	Positive affect related to attachment, desire, or expectation	Affection, Lust, Longing	Positive
Joy	Positive affect associated with pleasure, fulfillment, or elevated motivation	Cheerfulness, Zest, Contentment, Pride, Optimism, Enthrallment, Relief	Positive
Surprise	A sudden response to unexpected events or stimuli	Surprise	Mixed
Anger	Negative affect triggered by frustration, offense, aversion, or social comparison	Irritation, Exasperation, Rage, Disgust, Envy, Torment	Negative
Sadness	Negative affect related to loss, pain, disappointment, or social-emotional deprivation	Suffering, Sadness, Disappointment, Shame, Neglect, Sympathy	Negative
Fear	Negative affect driven by threat, uncertainty, or anticipated harm	Horror, Nervousness	Negative

Table 4: Data distribution of explicit and implicit emotions in MIE.

Emotion Category	Explicit	Implicit	Total
Neutral	68	–	68
Love	71	67	138
Joy	70	82	152
Surprise	70	41	111
Anger	69	78	147
Sadness	64	73	137
Fear	71	86	157
Total	483	427	910

balances psychological interpretability, annotation feasibility, and computational usability, making it an effective label system for multimodal explicit–implicit emotion understanding.

A.4 Detailed Data Distribution

Table 4 shows that the overall data distribution of MIE is relatively balanced, while still preserving the natural variability between explicit and implicit emotional expression. The explicit subset is highly uniform, with category sizes ranging only from 64 to 71, which helps reduce the influence of class imbalance on training and evaluation. This balanced design makes performance differences across explicit categories less likely to be driven by data quantity alone. In total, the explicit subset contains 483 samples.

Compared with the explicit subset, the implicit subset is moderately less balanced, with category sizes ranging from 41 to 86. Among the implicit categories, *Surprise* has the fewest samples (41), whereas *Fear* has the most (86), followed by *Joy* (82) and *Anger* (78). This distribution reflects the realistic difficulty of collecting and verifying different hidden emotions in multimodal scenes. In particular, implicit *Surprise* is relatively scarce, likely because its latent form is highly transient and often ambiguous in valence, making consistent annotation more difficult. By contrast, implicit *Fear*, *Anger*, and *Sadness* tend to leave more recoverable contextual or behavioral traces, which makes them easier to identify and annotate at scale.

Overall, MIE contains 910 samples, including 483 explicit and 427 implicit instances, indicating that the two expression forms are kept at a relatively comparable scale. This distribution is suitable for the unified explicit–implicit setting of MIE: it avoids severe data skew between the two expression types, while still preserving realistic category-level variation within the implicit subset. As a result, the dataset supports both stable quantitative evaluation and more fine-grained analysis of category-specific challenges in implicit emotion understanding.

A.5 Train/Validation/Test Split

MIE is split into training, validation, and test sets with a ratio of 7:1:2. The training set is used for parameter learning or prompt construction, the validation set for model selection and hyperparameter tuning, and the test set for final performance evaluation. During data splitting, we preserve the class distribution across subsets as much as possible to reduce the impact of class imbalance on evaluation results.

It is important to note that the goal of MIE is not merely to improve a model's ability to recognize implicit emotions, but also to equip the model with the ability to jointly distinguish explicit and implicit emotions. In real-world interactions, a model must first determine whether the current emotion is directly expressed or instead conveyed in a hidden, weakened, or indirect manner, before further identifying the specific emotion category. Therefore, implicit emotion samples are not treated as the sole learning target; instead, both explicit and implicit samples are incorporated into a unified label space for joint modeling.

Under this setting, the model is required to solve two coupled recognition problems: identifying the emotion category itself, and determining whether it is expressed explicitly or implicitly. This task formulation is more consistent with the requirements of realistic emotion understanding, and makes MIE a benchmark not only for implicit emotion inference, but also for the joint modeling of explicit and implicit emotions.

A.6 Why Extensive Context Is Not Used for Implicit Emotion Inference

Although contextual information is valuable for understanding implicit emotions, we do not use extensive context for implicit emotion inference because doing so would introduce a mismatch between the intended task setting and the actual reasoning process.

The goal of this work is to infer a character's latent emotional state from the current scene, rather than to solve the problem by relying on long-range narrative memory. If arbitrarily large amounts of preceding and following context were provided, the task could gradually shift from emotion understanding in the target scene to answer retrieval from the broader storyline. In that case, model performance might reflect access to external narrative clues rather than the ability to interpret subtle multimodal evidence within the scene itself.

A second reason is that extensive context may introduce substantial noise and dilute the emotionally relevant evidence. In realistic video narratives, long contextual windows often contain many events, speakers, and topic shifts that are only weakly related to the target emotional moment. For implicit emotion understanding, the key signals are often local and fine-grained, such as a restrained tone, a brief hesitation, a slight behavioral inconsistency, or a contextually meaningful utterance. Excessive context may obscure these crucial cues by overwhelming the model with irrelevant information, making it harder to identify which evidence is directly useful for the current inference. This problem is especially serious for large models, which may be tempted to over-explain the scene using broad narrative associations instead of grounding their decision in the actual multimodal traces of the target clip.

Finally, restricting the use of context helps preserve both task difficulty and evaluation validity. Implicit emotion understanding is challenging precisely because the true affective state is not fully revealed on the surface, and must be inferred from subtle evidence within a limited local interaction. If extensive context were freely available, some difficult cases could become artificially easier, not because the model has become better at understanding hidden emotions, but because the answer is indirectly exposed elsewhere in the narrative. Therefore, our design deliberately limits inference to the current scene and its immediate local evidence, while allowing contextual information only in a controlled manner during annotation. This setting ensures that the benchmark evaluates genuine implicit emotion reasoning rather than long-context story reconstruction, and makes the resulting performance gains more faithfully reflect improvements in multimodal affective inference.

A.7 Use of Contextual Information in the Annotation Process

n the annotation process, contextual information is introduced as an auxiliary reference to improve the reliability of emotion labeling, especially for implicit emotions. Unlike explicit emotions, which are often directly observable from facial expression, vocal prosody, or lexical content, implicit emotions are frequently weakly expressed, socially masked, or only partially reflected in the target clip itself. As a result, annotators may face substantial ambiguity if they rely on the isolated segment alone. Allowing access to surrounding context helps them better understand the local interactional situation, including event development, interpersonal relations, and conversational dynamics, which are often essential for interpreting the latent affective state.

At the same time, the use of context is carefully constrained to prevent the annotation process from becoming unrestricted narrative inference. The final label is always assigned to the target segment itself, and contextual information is used only to clarify the emotional meaning of that segment rather than to replace it with evidence from earlier or later scenes. In other words, context functions as a grounding signal for interpreting subtle multimodal cues, not as an external shortcut for directly determining the answer. This distinction is important because the goal of the dataset is still to evaluate emotion understanding from the current scene, while ensuring that human annotation remains as accurate and consistent as possible.

This controlled use of contextual information is particularly beneficial for implicit emotion annotation because the same surface behavior may correspond to different underlying emotions under different situational conditions. A pause, a restrained tone, a brief smile, or a neutral utterance may signal very different latent states depending on the surrounding interaction. By incorporating contextual information in a limited and principled way, the annotation process better captures the distinction between observable expression and underlying emotion, thereby reducing ambiguity, improving inter-annotator consistency, and enhancing the overall validity of the resulting dataset.

B Workflow Analysis and Additional Results

B.1 Prompt Templates for Each Component

All modules in *EmotionReasoner* are instantiated with the same backbone model. To improve output stability and facilitate cross-module interaction, we use structured prompt specifications instead of unconstrained free-form prompting. All modules operate under a shared emotion space, including seven explicit emotions (*Neutral, Love, Joy, Surprise, Anger, Sadness, Fear*) and six implicit emotions (*Love, Joy, Surprise, Anger, Sadness, Fear*), and are restricted to the current scene and the provided multimodal evidence only.

Task Controller. The *Task Controller* is responsible for high-level routing and final decision aggregation. In the routing stage, it determines whether the current scene can be recognized mainly from direct surface cues or requires deeper implicit reasoning. In the aggregation stage, it integrates the outputs of the other modules and produces the final prediction. Its prompt is formulated as follows:

Emotion Perceptor. The *Emotion Perceptor* extracts fine-grained affective cues from video, audio, and text, and provides an initial surface-level emotional impression. Its prompt emphasizes direct observation rather than deep interpretation, so that later reasoning modules can build on reliable low-level evidence.

Contextual Evidence Modeler. The *Contextual Evidence Modeler* organizes local contextual evidence and explains how the scene context constrains emotion interpretation. Instead of focusing on raw perceptual signals, this module structures the scene into event-level and interaction-level evidence units and highlights possible mismatches between outward behavior and inner state.

Implicit Emotion Reasoner. The *Implicit Emotion Reasoner* performs the final latent affective inference when the true emotional state is not directly expressed. It explicitly reasons over subtle cues, contextual constraints, and surface–latent discrepancy to determine the most likely implicit emotion.

Algorithm 1 Prompt Specification of Task Controller

Require: Video observations, audio observations, transcript/dialogue, scene description (if available), outputs of other modules (for final aggregation)
Ensure: Routing decision, candidate emotions, final emotion label, confidence, concise explanation

Summarize the overall emotional impression of the current scene
Determine whether the dominant emotion is *explicit*, *implicit*, *mixed*, or *uncertain*
Estimate whether direct surface multimodal cues are sufficient for reliable recognition
if surface cues are insufficient **then**
    Route the sample to deeper implicit reasoning
else
    Keep the sample in the direct explicit recognition path
end if
Produce 2–3 candidate emotions with confidence estimates
Integrate outputs from Emotion Perceptor, Contextual Evidence Modeler, and Implicit Emotion Reasoner
Compare surface evidence and latent emotion inference
Predict the final expression type and final emotion label
Output the final decision with supporting evidence and rejected alternatives

Algorithm 2 Prompt Specification of Emotion Perceptor

Require: Video observations, audio observations, transcript/dialogue, scene description (if available)
Ensure: Fine-grained multimodal affective cues, cross-modal consistency, candidate emotions

Extract directly observable **visual cues** (e.g., facial expression, gaze, posture, gesture, movement)
Extract directly observable **acoustic cues** (e.g., tone, intensity, pitch, pauses, trembling, laughter, sighs)
Extract directly observable **textual cues** (e.g., word choice, sentiment, contradiction, emphasis, indirect wording)
Determine whether the three modalities are consistent or inconsistent
Summarize the surface-level emotional impression of the current scene
Propose candidate emotions supported by direct evidence only
Highlight subtle or abnormal multimodal signals that may indicate hidden affective states
Avoid inferring deep hidden motives unless strongly grounded in the input

Structured Outputs. To ensure stable cross-module interaction, all modules are required to return structured outputs including candidate labels, confidence estimates, supporting evidence, and concise rationales. This design reduces free-form drift and makes intermediate reasoning results easier to inspect, compare, and aggregate.

Algorithm 3 Prompt Specification of Contextual Evidence Modeler

Require: Video observations, audio observations, transcript/dialogue, scene description, Emotion Perceptor output
Ensure: Structured contextual evidence, event units, interaction context, surface–latent mismatch description

Identify key events, interaction states, and speaker behaviors in the current scene
Infer local contextual factors relevant to emotion interpretation: interpersonal relation, conversational tension/harmony, implied goal or intention, possible social masking
Organize the scene into structured event-level and interaction-level evidence units
for each major candidate emotion **do**
    List supporting contextual evidence
    List contradictory contextual evidence
end for
Determine whether a mismatch exists between outward behavior and latent emotional state
Summarize why the emotion may be implicit rather than explicit
Avoid inventing long-range backstory unless explicitly provided

Algorithm 4 Prompt Specification of Implicit Emotion Reasoner

Require: Video observations, audio observations, transcript/dialogue, scene description, Emotion Perceptor output, Contextual Evidence Modeler output
Ensure: Final implicit emotion decision, candidate ranking, rationale, uncertainty note

Examine whether the surface emotional impression is aligned with the latent emotional state
if surface and latent emotions are misaligned **then**
    Explain the mismatch concisely
end if
Infer the most likely latent emotion from the implicit emotion set:
    {*Love, Joy, Surprise, Anger, Sadness, Fear*}
Rank the top implicit candidates with confidence estimates
Compare the main alternatives and explain why they are less plausible
Ground the decision in subtle multimodal cues, contextual constraints, behavioral inconsistency, and interactional meaning
Produce the final implicit emotion prediction with a concise rationale

B.2 Why Are Certain Implicit Emotion Categories More Difficult to Recognize?

Some implicit emotion categories are harder to recognize because their underlying affective states are often expressed through weak, indirect, and context-dependent signals rather than stable surface patterns. Unlike explicit emotions, which can often be identified from salient facial expressions, vocal prosody, or emotionally charged words, implicit emotions are frequently socially masked or only partially reflected in observable behavior. As a result, the

Table 5: Ablation study of Emotion Perceptor on explicit and implicit emotion understanding.

Setting	Explicit		Implicit	
	ACC ↑	WAF ↑	ACC ↑	WAF ↑
Full Model	84.68	79.67	54.80	60.99
w/o Global Emotion	83.02	78.42	53.86	58.26
w/o Fine-grained Cues	85.09	78.02	45.43	53.88

Table 6: Ablation study of Contextual Evidence Modeler on explicit and implicit emotion understanding.

Setting	Explicit		Implicit	
	ACC ↑	WAF ↑	ACC ↑	WAF ↑
Full Model	84.68	79.67	54.80	60.99
w/o Evidence Organizer	84.06	78.42	52.22	58.43
w/o Context Modeling	84.47	79.11	49.18	58.27

model must infer the latent emotional state from incomplete multimodal evidence, which makes recognition substantially more difficult. This difficulty is especially pronounced for categories such as *Fear*, *Anger*, and *Surprise*, whose hidden forms often lack clear and distinctive surface prototypes.

Another important reason is that the boundaries between some implicit emotion categories are inherently more ambiguous. Negative emotions such as *Anger*, *Sadness*, and *Fear* may share overlapping multimodal traces, including tension, hesitation, reduced verbal elaboration, or restrained expression, even though their underlying psychological states are different. Similarly, *Surprise* is often transient and ambiguous in valence, making it difficult to distinguish from *Joy* or *Fear* without sufficient contextual grounding. Therefore, the challenge is not simply detecting whether an emotion exists, but resolving fine-grained distinctions among semantically adjacent latent states.

Finally, the recognition difficulty of implicit categories is closely tied to how much interpretable evidence the hidden emotion leaves in the local interaction context. Some emotions, such as implicit *Love* or *Sadness*, may still leave relatively stable contextual traces through affiliative behavior or loss-related interaction patterns. In contrast, hidden *Fear* and *Anger* are more likely to appear as fragmented, non-prototypical, or socially regulated signals, which are harder to integrate reliably. This suggests that lower accuracy on certain implicit categories does not simply reflect weaker model performance, but rather the fact that these categories more strongly expose the core difficulty of implicit emotion understanding: recovering latent affective states from subtle, ambiguous, and context-dependent multimodal evidence.

B.3 Fine-Grained Ablation of EmotionReasoner Components

B.3.1 Fine-Grained Ablation of Emotion Perceptor. Table 5 provides a more fine-grained dissection of the *Emotion Perceptor* and shows that its two sub-functions—*global emotion perception* and *fine-grained cue extraction*—play different roles in the overall workflow. Removing *Global Emotion* causes moderate degradation on both explicit and implicit emotion understanding, indicating that a coarse global affective impression serves as a useful high-level prior for subsequent reasoning. This global signal likely helps the system maintain overall emotional coherence by anchoring local evidence to a plausible scene-level interpretation. However, its contribution remains relatively limited compared with that of *Fine-grained Cues*. Once fine-grained cues are removed, the model suffers only minor change in explicit ACC but a dramatic drop in implicit performance, with Imp-ACC and Imp-WAF decreasing by 9.37 and 7.11 points, respectively. This asymmetry is highly informative: it suggests that explicit emotion understanding can still be supported by relatively coarse multimodal impressions, whereas implicit emotion understanding depends much more critically on subtle, localized, and potentially non-prototypical affective traces.

More importantly, the results clarify *why* fine-grained cue extraction is so essential. Hidden emotions are rarely conveyed through strong and stable surface prototypes; instead, they often appear through weak signals such as restrained tone, slight hesitation, minor gaze shift, indirect wording, or local behavioral inconsistency. These signals may not be sufficient on their own, but they provide the crucial evidence from which later reasoning modules can infer a latent affective state. Without them, the model may still form a reasonable global impression of the scene, but it becomes much less capable of distinguishing whether the true emotion is only partially revealed, socially masked, or semantically displaced. In this sense, the Emotion Perceptor does more than summarize surface affect: it acts as the main interface through which weak multimodal irregularities are preserved and exposed to the rest of the workflow. Overall, the ablation shows that global emotion perception contributes useful scene-level guidance, while fine-grained cue extraction is the more decisive factor that enables EmotionReasoner to move beyond surface recognition and achieve strong implicit emotion understanding.

B.3.2 Fine-Grained Ablation of Contextual Evidence Modeler. Table 6 provides a more fine-grained analysis of the *Contextual Evidence Modeler*, showing that its two sub-functions—*evidence organization* and *context modeling*—contribute to implicit emotion understanding in different but complementary ways. Removing the *Evidence Organizer* leads to a moderate decline on both explicit and implicit emotion recognition, with particularly noticeable drops in Imp-ACC and Imp-WAF. This suggests that explicitly structuring multimodal clues into coherent evidence units is useful for stabilizing downstream reasoning, since implicit emotions often rely on several weak signals that are only informative when considered together. Without such organization, the model still has access to the same raw cues, but it becomes less effective at integrating them into a consistent latent-state hypothesis. In contrast, removing *Context Modeling* causes a much larger drop in implicit ACC while leaving explicit performance almost unchanged, indicating that local context is especially critical for recovering hidden emotions whose meaning cannot be determined from surface behavior alone.

More importantly, the different ablation patterns reveal the distinct roles of these two components. The *Evidence Organizer* mainly improves the *structural coherence* of reasoning by converting fragmented multimodal observations into interpretable evidence units, which helps preserve useful weak signals for later inference. The *Context Modeling* component, however, contributes more directly

Table 7: Ablation study of Implicit Emotion Reasoner on explicit and implicit emotion understanding.

Setting	Explicit		Implicit	
	ACC ↑	WAF ↑	ACC ↑	WAF ↑
Full Model	84.68	79.67	54.80	60.99
w/o Implicit Emotion Inference	84.47	77.32	47.30	52.16
w/o Explanation Generation	84.88	78.01	54.10	59.37

to *latent emotion disambiguation*: it situates these evidence units within interaction dynamics, conversational tension, and possible social masking, enabling the model to explain why a given behavior may reflect an implicit rather than explicit emotional state. This explains why removing context modeling produces the largest drop in Imp-ACC: once contextual grounding is weakened, the model can still perceive emotional irregularities, but struggles to infer which hidden affective state they actually indicate. Overall, the ablation suggests that evidence organization helps the model retain and combine subtle cues, while context modeling gives those cues emotional meaning; their combination is therefore essential for robust implicit emotion understanding.

B.3.3 Fine-Grained Ablation of Implicit Emotion Reasoner. Table 7 provides a more fine-grained analysis of the *Implicit Emotion Reasoner*, showing that its two sub-functions—*implicit emotion inference* and *explanation generation*—make different contributions to the overall workflow. Removing *Implicit Emotion Inference* causes a substantial degradation on implicit emotion recognition, with Imp-ACC and Imp-WAF dropping by 7.50 and 8.83 points, respectively, while explicit ACC changes only marginally. This pattern clearly indicates that the core value of the Implicit Emotion Reasoner lies in its ability to transform perceptual and contextual evidence into a latent-state decision, rather than merely refining already visible emotional cues. Once this inference function is removed, the model may still preserve a reasonable surface-level understanding of the scene, but it becomes much less capable of deciding which hidden emotion best explains the observed multimodal irregularities. By contrast, removing *Explanation Generation* leads to a relatively smaller drop, suggesting that generating an explicit verbal rationale is not the primary source of performance gains, although it still provides measurable benefits for both explicit and implicit emotion understanding.

More importantly, the different ablation patterns reveal that these two sub-functions operate at different levels. *Implicit Emotion Inference* is the main decision-making component: it resolves the mismatch between outward expression and underlying affective state, ranks plausible latent candidates, and determines which implicit category is most consistent with the available evidence. This is why its removal sharply weakens implicit performance, especially on WAF, where balanced recognition across categories requires stronger disambiguation. In contrast, *Explanation Generation* appears to play a supporting but still meaningful role. Producing an explicit explanation likely encourages the model to maintain a more structured and evidence-grounded reasoning process, reducing arbitrary predictions and improving decision consistency. In other words, explanation generation does not replace latent inference, but helps regularize it. Overall, the ablation suggests that the Implicit Emotion Reasoner improves implicit emotion understanding primarily through explicit latent-state inference, while explanation generation further stabilizes this process by encouraging more coherent and interpretable reasoning.

B.4 Why Does the Proposed Workflow Improve Implicit Emotion Understanding?

The proposed emotion understanding workflow improves implicit emotion understanding because it is better aligned with the intrinsic difficulty of the task. Unlike explicit emotions, implicit emotions are often weakly expressed, socially masked, or only partially reflected in facial expression, vocal prosody, and literal text. As a result, direct end-to-end prediction tends to over-rely on the most salient surface cues and may fail when outward behavior does not fully reveal the true affective state. In this sense, implicit emotion understanding is not simply a perception problem, but a reasoning problem that requires the model to go beyond surface-level observations.

EmotionReasoner addresses this challenge by decomposing the task into several complementary stages. The Emotion Perceptor first extracts fine-grained multimodal affective cues, helping the system capture subtle signals that may otherwise be overlooked. The Contextual Evidence Modeler then organizes local contextual evidence, such as interaction dynamics, conversational tension, and possible social masking, so that emotion interpretation is grounded in the scene rather than isolated cues alone. Based on these intermediate results, the Implicit Emotion Reasoner further infers the latent emotional state by explicitly considering the possible mismatch between surface expression and underlying emotion.

Finally, the Task Controller integrates the outputs of all modules to produce a more balanced final decision. Therefore, the advantage of the proposed workflow does not come from simply adding more processing steps, but from enabling progressive evidence accumulation, context-aware interpretation, and latent-state reasoning in a structured manner. This is also consistent with the experimental results: the gains of EmotionReasoner are concentrated on implicit emotions, where stronger perception alone is insufficient, while structured evidence organization and reasoning become essential for reliable recognition.

B.5 Common Failure Modes of MLLMs in Implicit Emotion Recognition

B.5.1 Qwen2.5-Omni-7B. As shown in Figure 1, the confusion matrix of Qwen2.5-Omni-7B reveals a clear asymmetry between explicit and implicit emotion recognition. For explicit emotions, the model already forms a relatively sharp diagonal, indicating that it can rely on facial expression, vocal prosody, and literal textual content to recognize directly expressed emotions with reasonable accuracy. In contrast, the diagonal becomes much weaker and the error distribution becomes much more dispersed for implicit emotions, showing that the model struggles once the true affective state is hidden, weakened, or only indirectly expressed. This suggests that the main limitation of Qwen2.5-Omni-7B does not lie in basic multimodal perception itself, but in its inability to recover latent emotional states from subtle anomalies and local contextual evidence.

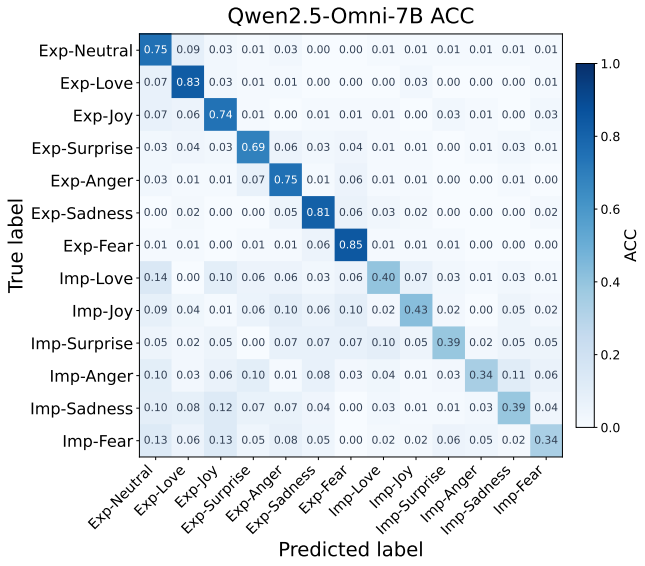


Figure 1: Row-normalized confusion matrix of *Qwen2.5-Omni-7B* on the unified 13-class label space.

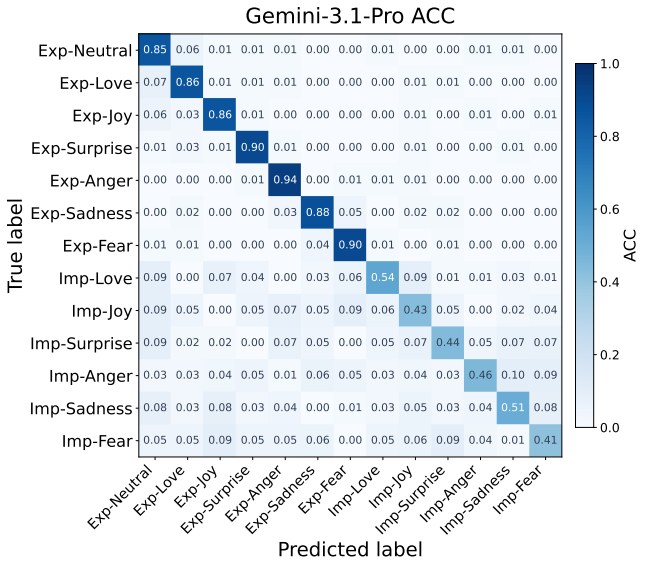


Figure 2: Row-normalized confusion matrix of *Gemini-3.1-Pro* on the unified 13-class label space.

A more specific failure pattern is that the model tends to overrely on surface cues and drift from implicit labels to explicit ones. As shown in the figure, a considerable portion of implicit samples are misclassified into explicit categories, rather than remaining within the implicit subspace. This indicates that when the latent emotion leaves only weak, fragmented, or incomplete evidence, Qwen2.5-Omni-7B often does not further examine whether the outward expression is consistent with the true inner state, but instead defaults to the most salient visual, acoustic, or lexical cue available in the current scene. In other words, the model can often detect the emotional surface of the interaction, but fails to explain what hidden affective state that surface behavior actually implies.

At the same time, the implicit subspace also exhibits substantial confusion among semantically adjacent categories, especially for emotions with overlapping hidden traces. This suggests that Qwen2.5-Omni-7B is not entirely incapable of sensing emotional irregularity; rather, it often reaches only a coarse judgment that "something emotionally non-typical is happening," without being able to determine which latent emotion best explains the observed behavior. This problem is particularly severe for negative emotions, whose implicit forms often share restrained tone, hesitation, emotional suppression, and context-dependent behavioral irregularities. Overall, the figure does not indicate random classification errors, but rather systematic limitations of Qwen2.5-Omni-7B in weak-signal integration, context-sensitive interpretation, and latent emotion disambiguation, which are exactly the core challenges of implicit emotion understanding.

B.5.2 Gemini-3.1-Pro. As shown in Figure 2, the confusion matrix of Gemini-3.1-Pro reveals a similar but somewhat stronger explicit–implicit asymmetry than that of other strong baselines. On the explicit side, the matrix exhibits a very sharp diagonal, with most explicit categories reaching self-recall values around 0.85–0.94, such as Exp-Anger = 0.94, Exp-Surprise = 0.90, and Exp-Fear = 0.90. This indicates that Gemini-3.1-Pro is highly effective at recognizing directly expressed emotions from salient multimodal cues. However, its performance drops substantially once the task shifts to implicit emotions, whose diagonal values fall to roughly 0.41–0.54, including Imp-Love = 0.54, Imp-Sadness = 0.51, Imp-Anger = 0.46, and Imp-Fear = 0.41. This pattern suggests that Gemini-3.1-Pro has a very strong perceptual capacity for overt affective signals, but still struggles to recover latent emotional states when the true emotion is only weakly expressed, socially masked, or behaviorally indirect.

A more specific failure pattern is that Gemini-3.1-Pro still tends to drift from implicit labels toward explicit ones, although this tendency appears somewhat more controlled than in weaker baselines. As shown in the figure, many implicit samples are still misclassified into explicit categories, especially those with semantically or affectively related surface forms. For example, Imp-Love is confused with Exp-Neutral (0.09), Exp-Joy (0.07), and Exp-Fear (0.06); Imp-Joy is distributed across Exp-Neutral (0.09), Exp-Anger (0.07), Exp-Fear (0.09), and several other explicit categories; and Imp-Surprise is frequently confused with Exp-Neutral (0.09), Exp-Anger (0.07), and Exp-Sadness (0.05). These errors indicate that even though Gemini-3.1-Pro can detect that a scene carries emotional information, it still often defaults to the most accessible surface-level affective impression when the latent state is not fully observable. In other words, its strong multimodal perception does not automatically translate into equally strong latent-state inference.

At the same time, the implicit subspace itself remains notably entangled, especially for semantically adjacent or weakly signaled

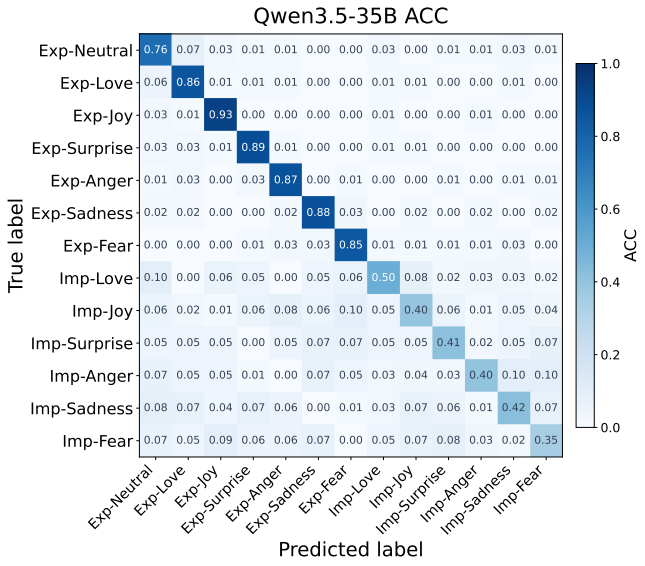


Figure 3: Row-normalized confusion matrix of *Qwen3.5-35B* on the unified 13-class label space.

emotions. This is particularly visible in categories such as Imp-Anger, Imp-Sadness, and Imp-Fear, which still show substantial mutual confusion, for example Imp-Anger to Imp-Sadness (0.10) and Imp-Anger to Imp-Fear (0.09), as well as broader dispersion of Imp-Fear across both explicit and implicit labels. Such patterns suggest that Gemini-3.1-Pro is often able to recognize that the target emotion is non-explicit, but still struggles to determine which latent affective state best explains the observed multimodal irregularities. Overall, the figure indicates that the main limitation of Gemini-3.1-Pro in implicit emotion understanding is not failure of basic perception, but insufficient weak-signal integration, context-sensitive interpretation, and fine-grained disambiguation among hidden emotions. This also explains why Gemini-3.1-Pro achieves very strong explicit results yet still leaves clear room for improvement on the implicit subset.

B.5.3 Qwen3.5-35B. As shown in Figure 3, the confusion matrix of Qwen3.5-35B reveals a clear gap between explicit and implicit emotion recognition. On the explicit side, the model already forms a strong diagonal structure, with most explicit categories reaching self-recall values around 0.76–0.93, such as Exp-Joy = 0.93, Exp-Surprise = 0.89, Exp-Sadness = 0.88, and Exp-Anger = 0.87. This indicates that Qwen3.5-35B can reliably recognize emotions that are directly expressed through salient multimodal cues. In contrast, its implicit categories remain much weaker, with diagonal values mostly around 0.35–0.50, including Imp-Love = 0.50, Imp-Joy = 0.40, Imp-Surprise = 0.41, Imp-Anger = 0.40, Imp-Sadness = 0.42, and Imp-Fear = 0.35. This pattern suggests that, despite its strong capability on explicit affective perception, Qwen3.5-35B still struggles when the underlying emotional state is only partially observable and must be inferred from weak, indirect, or context-dependent evidence.

A more specific failure pattern is that Qwen3.5-35B tends to fallback from implicit labels to explicit ones when latent evidence is insufficiently grounded. As shown in the figure, many implicit samples are dispersed across explicit categories rather than remaining within the implicit subspace. For instance, Imp-Love is confused with Exp-Neutral (0.10), Exp-Joy (0.06), and Exp-Fear (0.06); Imp-Joy drifts toward Exp-Anger (0.08), Exp-Fear (0.10), Exp-Sadness (0.06), and Exp-Surprise (0.06); and Imp-Fear is broadly distributed across several explicit categories, including Exp-Joy (0.09), Exp-Sadness (0.07), Exp-Surprise (0.06), and Exp-Anger (0.06). These error patterns indicate that when the true emotion is not clearly exposed, Qwen3.5-35B often defaults to the most accessible surface-level emotional impression rather than reasoning about whether the observable behavior is masking a different latent state. In other words, its strong multimodal recognition ability does not automatically extend to robust hidden-emotion inference.

At the same time, the implicit subspace itself remains substantially entangled, especially for semantically neighboring negative emotions. Categories such as Imp-Anger, Imp-Sadness, and Imp-Fear exhibit persistent mutual confusion, for example Imp-Anger to Imp-Sadness (0.10), Imp-Anger to Imp-Fear (0.10), and broader dispersion of Imp-Fear into both explicit and implicit labels. This suggests that Qwen3.5-35B is often able to detect that a scene contains emotionally non-explicit or anomalous signals, but still struggles to determine which latent affective state best explains them. Overall, the figure shows that the main weakness of Qwen3.5-35B in implicit emotion recognition is not failure of basic multimodal perception, but insufficient weak-signal integration, context-sensitive interpretation, and fine-grained disambiguation among hidden emotions. This also explains why the model performs strongly on explicit emotions while still leaving substantial room for improvement on the implicit subset.

B.6 Correlation Between the Proposed Workflow and Human Emotional Judgment

The correlation with human emotional judgment provides an important perspective on the validity of the proposed workflow. Beyond achieving stronger benchmark performance, a reliable implicit emotion understanding framework should also produce decisions that are consistent with how humans interpret latent affective states in realistic multimodal interactions. This point is especially important for implicit emotions, where the target state is often weakly expressed, socially masked, or only partially reflected in observable behavior. In such cases, purely surface-driven prediction can easily diverge from human interpretation, whereas stronger alignment with human judgment suggests that the model is capturing not only the visible emotional surface, but also the underlying affective meaning that human annotators aim to infer.

The observed correlation can be understood from the design of EmotionReasoner itself. Human observers rarely infer hidden emotions from a single cue alone; instead, they integrate facial expression, vocal tone, lexical content, local interaction context, and possible discrepancies between outward behavior and inner state. The proposed workflow follows a similar logic. The Emotion Perceptor extracts fine-grained multimodal cues, the Contextual Evidence Modeler organizes local situational evidence, and the

Implicit Emotion Reasoner further infers the latent affective state by explicitly considering surface–latent mismatch. This structured process makes the workflow more compatible with human emotional reasoning than direct end-to-end prediction, which often over-relies on the most salient surface signals.

Overall, the correlation with human judgment further supports the effectiveness of the proposed workflow for implicit emotion understanding. Since implicit emotions are inherently ambiguous and context-dependent, agreement with human interpretation is itself an important indicator of practical usefulness. In this sense, the advantage of EmotionReasoner is not only quantitative but also cognitive: it improves performance by making the inference process more consistent with the way humans interpret concealed emotional states. This result therefore strengthens the claim that the proposed workflow does not merely fit dataset-specific patterns, but captures a more general and human-relevant mechanism for multimodal implicit emotion understanding.

C Ethics and Responsibility Statement

C.1 Dataset Release Statement

The MIE dataset is scheduled to be released after the completion of the overall project, subject to the necessary compliance review and copyright considerations. The released resources will include the sample split files, emotion annotations, and the associated metadata required for benchmarking and evaluation. Depending on copyright and platform constraints, the public version may consist of processed clips or, when redistribution is not permitted, only clip identifiers, timestamps, subtitles, and annotation files.

The dataset will be released strictly for academic research and educational use. It will not be intended for commercial use, identity inference, surveillance, or other high-stakes decision-making scenarios involving real individuals. Moreover, since implicit emotion inference is inherently uncertain and context-dependent, the dataset should not be treated as a tool for definitive judgment of a person's internal psychological state.

C.2 Legal Statement on Data Collection

The MIE dataset is constructed from film and television materials that are publicly distributed through authorized media channels. The data are collected solely for academic research on multimodal emotion understanding. To the best of our knowledge, the dataset construction process follows the principles of limited, research-oriented use and does not aim to reproduce or redistribute complete original works.

To reduce potential copyright and usage risks, we [only retain short clips necessary for analysis / avoid distributing raw video content whenever redistribution is not permitted / release only derived annotations and indexing information when required]. The dataset does not contain private recordings, hidden-camera materials, or personally submitted user data. All samples are derived from scripted audiovisual content performed by professional actors in public media productions. If any copyright holder or platform identifies a compliance concern, we will promptly review the relevant material and make necessary adjustments, including removal or restricted access when appropriate.

C.3 Annotator Statement

All annotations in MIE were completed by trained annotators under a unified annotation protocol. Before formal annotation, annotators were provided with written guidelines describing the emotion categories, the distinction between explicit and implicit emotions, and the role of contextual information in verification. Pilot annotation and discussion sessions were conducted to improve consistency before large-scale labeling.

During annotation, each sample was independently reviewed by multiple annotators, and disagreements were resolved through re-checking and discussion according to the predefined criteria. Annotators were instructed to focus on emotion understanding from multimodal evidence rather than personal preference or speculative interpretation. Since the dataset is based on public film and television content rather than private user data, the annotation process does not involve the collection of sensitive personal information from real-world participants. We acknowledge that implicit emotion annotation still contains a degree of subjectivity, and therefore treat the labels as carefully verified research annotations rather than absolute ground truth of internal mental states.

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